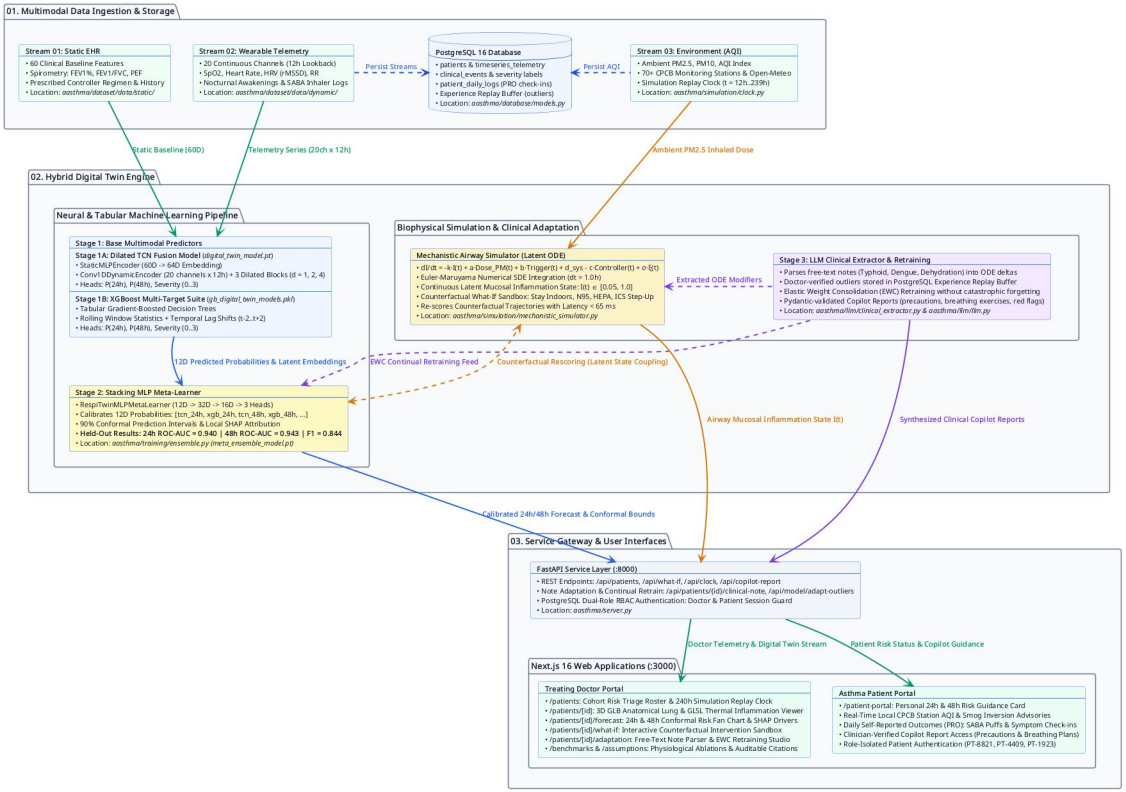

01. Multimodal Data Ingestion & Storage
Stream 01: Static EHR
• 60 Clinical Baseline Features
• Spirometry: FEV1%, FEV1/FVC, PEF
• Prescribed Controller Regimen & History
• Location: aasthma/dataset/data/static/
Stream 02: Wearable Telemetry
• 20 Continuous Channels (12h Lookback)
• SpO2, Heart Rate, HRV (rMSSD), RR
• Nocturnal Awakenings & SABA Inhaler Logs
• Location: aasthma/dataset/data/dynamic/
Persist Streams
PostgreSQL 16 Database
• patients & timeseries_telemetry
• clinical_events & severity labels
• patient_daily_logs (PRO check-ins)
• Experience Replay Buffer (outliers)
• Location: aasthma/database/models.py
Persist AQI
Stream 03: Environment (AQI)
• Ambient PM2.5, PM10, AQI Index
• 70+ CPCB Monitoring Stations & Open-Meteo
• Simulation Replay Clock (t = 12h..239h)
• Location: aasthma/simulation/clock.py
Static Baseline (60D)
Telemetry Series (20ch x 12h)
Ambient PM2.5 Inhaled Dose
02. Hybrid Digital Twin Engine
Neural & Tabular Machine Learning Pipeline
Stage 1: Base Multimodal Predictors
Stage 1A: Dilated TCN Fusion Model (digital_twin_model.pt)
• StaticMLPEncoder (60D -> 64D Embedding)
• ConvTDDynamicEncoder (20 channels x 12h) + 3 Dilated Blocks (d = 1, 2, 4)
• Heads: P(24h), P(48h), Severity (0..3)
Stage 1B: XGBoost Multi-Target Suite (gb_digital_twin_models.pkl)
• Tabular Gradient-Boosted Decision Trees
• Rolling Window Statistics + Temporal Lag Shifts (t-2..t+2)
• Heads: P(24h), P(48h), Severity (0..3)
Biophysical Simulation & Clinical Adaptation
Mechanistic Airway Simulator (Latent ODE)
• dI/dt = -k·I(t) + a·Dose_PM(t) + b·Trigger(t) + d_sys - c·Controller(t) + σ·ξ(t)
• Euler-Maruyama Numerical SDE Integration (dt = 1.0h)
• Continuous Latent Mucosal Inflammation State: I(t) ∈ [0.05, 1.0]
• Counterfactual What-If Sandbox: Stay Indoors, N95, HEPA, ICS Step-Up
• Re-scores Counterfactual Trajectories with Latency < 65 ms
• Location: aasthma/simulation/mechanistic_simulator.py
Extracted ODE Modifiers
Stage 3: LLM Clinical Extractor & Retraining
• Parses free-text notes (Typhoid, Dengue, Dehydration) into ODE deltas
• Doctor-verified outliers stored in PostgreSQL Experience Replay Buffer
• Elastic Weight Consolidation (EWC) Retraining without catastrophic forgetting
• Pydantic-validated Copilot Reports (precautions, breathing exercises, red flags)
• Location: aasthma/llm/clinical_extractor.py & aasthma/llm/llm.py
12D Predicted Probabilities & Latent Embeddings
EWC Continual Retraining Feed
Counterfactual Rescoring (Latent State Coupling)
Stage 2: Stacking MLP Meta-Learner
• Resp/TwinMLPMetaLearner (12D -> 32D -> 16D -> 3 Heads)
• Calibrates 12D Probabilities: [tcn_24h, xgb_24h, tcn_48h, xgb_48h, ...]
• 90% Conformal Prediction Intervals & Local SHAP Attribution
• Held-Out Results: 24h ROC-AUC = 0.940 | 48h ROC-AUC = 0.943 | F1 = 0.844
• Location: aasthma/training/ensemble.py (meta_ensemble_model.pt)
Airway Mucosal Inflammation State I(t)
Synthesized Clinical Copilot Reports
Calibrated 24h/48h Forecast & Conformal Bounds
03. Service Gateway & User Interfaces
FastAPI Service Layer (:8000)
• REST Endpoints: /api/patients, /api/what-if, /api/clock, /api/copilot-report
• Note Adaptation & Continual Retrain: /api/patients/{id}/clinical-note, /api/model/adapt-outliers
• PostgreSQL Dual-Role RBAC Authentication: Doctor & Patient Session Guard
• Location: aasthma/server.py
Doctor Telemetry & Digital Twin Stream
Patient Risk Status & Copilot Guidance
Next.js 16 Web Applications (:3000)
Treating Doctor Portal
• /patients: Cohort Risk Triage Roster & 240h Simulation Replay Clock
• /patients/[id]: 3D GLB Anatomical Lung & GLSL Thermal Inflammation Viewer
• /patients/[id]/forecast: 24h & 48h Conformal Risk Fan Chart & SHAP Drivers
• /patients/[id]/what-if: Interactive Counterfactual Intervention Sandbox
• /patients/[id]/adaptation: Free-Text Note Parser & EWC Retraining Studio
• /benchmarks & /assumptions: Physiological Ablations & Auditable Citations
Asthma Patient Portal
• /patient-portal: Personal 24h & 48h Risk Guidance Card
• Real-Time Local CPCB Station AQI & Smog Inversion Advisories
• Daily Self-Reported Outcomes (PRO): SABA Puffs & Symptom Check-ins
• Clinician-Verified Copilot Report Access (Precautions & Breathing Plans)
• Role-Isolated Patient Authentication (PT-8821, PT-4409, PT-1923)